

Problem Set: The Interpretation of Propositional Formulas

Goldrei §2.3, with emphasis on Theorem 2.2

Throughline. Theorem 2.2 has two halves. Uniqueness says a truth assignment has no freedom beyond its values on the propositional variables; existence says that on the variables the freedom is total. Much of this sheet is about knowing which half you are using at any given moment — and recognizing the situations in which neither is available.

Ground rules

- Notation and conventions follow Goldrei, Chapter 2: formulas are fully bracketed; the *length* of a formula is the number of occurrences of connectives in it; for a function $v: P \rightarrow \{T, F\}$, $\bar{v}$ denotes the unique truth assignment extending v given by Theorem 2.2. As in the book, “all truth assignments” abbreviates “all truth assignments defined on a set of propositional variables including those appearing in the formulas at hand.”
- You may use without proof: Theorem 2.2, anything established in the book up to the end of §2.3, and any earlier problem on this sheet.
- **Refutation discipline.** To refute a claim of the form “for all formulas $\varphi, \psi, \dots$ ” you must produce a fully concrete instance: (i) name a specific formula for each metavariable; (ii) verify that your formulas satisfy *every* hypothesis of the claim; (iii) exhibit a specific truth assignment, by listing its values on the relevant propositional variables, and compute the truth values needed to show that the conclusion fails. An explanation of why a counterexample ought to exist is not a counterexample.
- Starred problems are harder.

Part I — Reading Theorem 2.2

Problem 1 (Reading the theorem precisely). Let P be a set of propositional variables and $S = \{\neg, \wedge, \vee, \rightarrow, \leftrightarrow\}$. For each statement below, decide whether it is true or false. Justify each verdict briefly; for false statements, follow the refutation discipline.

- (a) For every function $v: P \rightarrow \{T, F\}$ there is *at least one* truth assignment w with $w(p) = v(p)$ for all $p \in P$.
- (b) For every function $v: P \rightarrow \{T, F\}$ there is *at most one* truth assignment w with $w(p) = v(p)$ for all $p \in P$.
- (c) For every function $v: P \rightarrow \{T, F\}$ there is exactly one *function* $w: \text{Form}(P, S) \rightarrow \{T, F\}$ with $w(p) = v(p)$ for all $p \in P$.
- (d) For every truth assignment $w: \text{Form}(P, S) \rightarrow \{T, F\}$ there is exactly one function $v: P \rightarrow \{T, F\}$ such that $w = \bar{v}$.
- (e) If u, w are truth assignments with $u(p) = w(p)$ for all $p \in P$, then $u(\varphi) = w(\varphi)$ for all $\varphi \in \text{Form}(P, S)$.

- (f) If u is a truth assignment, $w: \text{Form}(P, S) \rightarrow \{T, F\}$ is a function, and $u(p) = w(p)$ for all $p \in P$, then w is a truth assignment.
- (g) If u, w are truth assignments with $u(\varphi) = w(\varphi)$ for every formula φ of length at least 1, then $u = w$.

Problem 2 (Extensions that are not truth assignments). Let $P = \{p\}$, $S = \{\neg, \wedge, \vee, \rightarrow, \leftrightarrow\}$, and let $v: P \rightarrow \{T, F\}$ be given by $v(p) = T$.

- (a) Exhibit two distinct functions $f, g: \text{Form}(P, S) \rightarrow \{T, F\}$ with $f(p) = g(p) = T$, neither of which is a truth assignment. For each, name a specific formula at which it violates a specific truth-table clause, and verify the violation.
- (b) Show that there are infinitely many functions $\text{Form}(P, S) \rightarrow \{T, F\}$ extending v .
- (c) In light of (b), exactly how many truth assignments extend v ? Which result are you citing?
- (d) Prove that restriction to P and the map $v \mapsto \bar{v}$ are mutually inverse bijections between the set of truth assignments on $\text{Form}(P, S)$ and the set of functions $P \rightarrow \{T, F\}$. Deduce that if $|P| = n$ then there are exactly 2^n truth assignments on $\text{Form}(P, S)$ (cf. Exercise 2.19).

Problem 3 (What unique readability is for). Work with strings over the alphabet $\{p, q, r, \wedge, \vee\}$ and *no brackets*. Define the *pseudo-formulas* to be the smallest class of strings such that each of p, q, r is a pseudo-formula, and whenever α, β are pseudo-formulas, so are the strings $\alpha \wedge \beta$ and $\alpha \vee \beta$. Consider the string σ given by $p \wedge q \vee r$.

- (a) Exhibit two distinct construction trees for σ .
- (b) Let $v(p) = F$, $v(q) = v(r) = T$. Compute a truth value for σ bottom-up along each of your two trees, using the usual truth tables for $\wedge$ and $\vee$. Conclude that the recursive clauses do *not* determine a function from pseudo-formulas to $\{T, F\}$.
- (c) Identify the sentence in the existence half of Goldrei's proof of Theorem 2.2 that fails for pseudo-formulas, and explain exactly what goes wrong with it here.
- (d) Consider the attempted definition " $\bar{v}(\sigma) :=$ the value computed from the decomposition of σ ." Say precisely which of *existence*, *uniqueness*, and *well-definedness* of $\bar{v}$ is threatened by (b), and explain how these three notions are related in this situation.

Problem 4 (Adding a connective). Let $\oplus$ ("exclusive or") be a new binary connective whose truth table is

$v(\theta)$	$v(\psi)$	$v((\theta \oplus \psi))$
T	T	F
T	F	T
F	T	T
F	F	F

and let $S_{\oplus} = \{\neg, \wedge, \vee, \rightarrow, \leftrightarrow, \oplus\}$.

- (a) Write out the clause " v respects the truth table of $\oplus$ " in the style of §2.3.
- (b) State the analogue of Theorem 2.2 for the language $\text{Form}(P, S_{\oplus})$.
- (c) Describe exactly what must be added or changed in each half of the proof — the existence half and the uniqueness half — to prove your statement from (b).

- (d) The proof relies on facts about $\text{Form}(P, S_{\oplus})$ of the kind established for $\text{Form}(P, S)$ in §2.2. Identify these facts precisely, and explain what would go wrong in your argument from (c) if the uniqueness of the principal connective failed for the enlarged language.

Part II — Computing with truth assignments

Problem 5 (Bottom-up evaluation). Let w be a truth assignment with $w(p) = \text{F}$, $w(q) = \text{T}$, $w(r) = \text{T}$.

- (a) Compute $w(((\neg p \vee q) \rightarrow (r \wedge \neg q)))$, listing the value of every subformula.
 (b) Compute $w(((p \rightarrow (q \rightarrow r)) \leftrightarrow \neg(p \vee \neg r)))$, listing the value of every subformula.

For (c) and (d), say that the information above *determines* $w(\varphi)$ if every truth assignment w satisfying $w(p) = \text{F}$, $w(q) = \text{T}$, $w(r) = \text{T}$ gives φ the same value. (Note that nothing has been said about $w(s)$.)

- (c) Is $w(((q \wedge s) \vee r))$ determined? If so, compute it and justify; if not, exhibit two truth assignments consistent with the information above that give the formula different values.
 (d) Same question for $w(((q \wedge s) \wedge r))$.
 (e) Now suppose all you know is $w(p) = \text{T}$. For each of $(p \vee q)$, $(p \wedge q)$, $(q \rightarrow p)$, $(q \vee \neg q)$, $(q \wedge \neg q)$, decide whether $w(\varphi)$ is determined, and justify as in (c) and (d).

Problem 6 (Truth tables, concrete and schematic). (a) Give the full truth table of $((p \vee \neg q) \rightarrow (q \wedge r))$.

- (b) Give the truth table of $((\varphi \rightarrow \psi) \rightarrow \varphi)$ in terms of the truth values of the formulas φ and ψ , as in Exercise 2.23.
 (c) Realize the row $(\varphi, \psi) = (\text{F}, \text{T})$ of your table concretely: produce specific formulas φ_0, ψ_0 built from p, q and a truth assignment w (give its values on p and q) with $w(\varphi_0) = \text{F}$ and $w(\psi_0) = \text{T}$, then confirm by computation that $w(((\varphi_0 \rightarrow \psi_0) \rightarrow \varphi_0))$ agrees with your table.
 (d) Now take ψ to be the formula $\neg\varphi$. Which rows of your table in (b) remain realizable, and why do the others drop out? Show that $((\varphi \rightarrow \neg\varphi) \rightarrow \varphi)$ is a tautology if and only if φ is a tautology, and a contradiction if and only if φ is a contradiction.
 (e) Using truth tables or direct argument, classify each of the following as a tautology, a contradiction, or neither:

$$((p \rightarrow q) \vee (q \rightarrow p)), \quad (p \leftrightarrow \neg p), \quad ((p \vee q) \rightarrow p), \quad (\neg(p \wedge q) \leftrightarrow (\neg p \vee \neg q)).$$

Problem 7 (Counting assignments). Let $n \geq 3$ and work with truth assignments on $\{p_1, \dots, p_n\}$. How many of the 2^n truth assignments make each of the following formulas true?

- (a) $(p_1 \wedge \neg p_2)$
 (b) $\neg((p_1 \vee p_2) \vee p_3)$
 (c) $\neg(p_1 \rightarrow p_2)$
 (d) $((p_1 \vee p_2) \wedge p_3)$, in the case $n = 3$.

Justify each count.

Part III — Inductions on the length of a formula

Problem 8 (The relevance lemma). (a) Prove: if u, w are truth assignments and $\varphi \in \text{Form}(P, S)$, $S = \{\neg, \wedge, \vee, \rightarrow, \leftrightarrow\}$, is a formula such that $u(q) = w(q)$ for every propositional variable q occurring in φ , then $u(\varphi) = w(\varphi)$. Use induction on the length of φ ; write the base case and the cases $\neg, \wedge, \rightarrow$ in full, declaring the remaining cases similar. Make your induction hypothesis explicit, and flag the exact steps at which you use that u and w respect the truth tables.

- (b) Deduce the statement of Exercise 2.18, together with its analogue for the full connective set S .
- (c) Deduce that the truth-table method is sound: if the propositional variables occurring in φ are among $p_1, \dots, p_n$, then φ is a tautology if and only if every one of the 2^n truth assignments on $\text{Form}(\{p_1, \dots, p_n\}, S)$ makes φ true. Be careful: a truth assignment on $\{p_1, \dots, p_n\}$ and a truth assignment on a larger set P of variables are functions with different domains, and part of your task is to bridge the two.

Problem 9 (Negating the variables: Exercise 2.28 extended). For a truth assignment v , let v^* be the truth assignment determined, via Theorem 2.2, by

$$v^*(p) = \begin{cases} \text{T}, & \text{if } v(p) = \text{F}, \\ \text{F}, & \text{if } v(p) = \text{T}, \end{cases} \quad \text{for all propositional variables } p.$$

For $\varphi \in \text{Form}(P, S)$, $S = \{\neg, \wedge, \vee, \rightarrow, \leftrightarrow\}$, define φ^* by recursion: $p^* = \neg p$ for each variable p ; $(\neg\theta)^* = \neg\theta^*$; and $(\theta \circ \psi)^* = (\theta^* \circ \psi^*)$ for each $\circ \in \{\wedge, \vee, \rightarrow, \leftrightarrow\}$. (As always, this definition is unambiguous thanks to unique readability.)

- (a) Prove that $v(\varphi) = v^*(\varphi^*)$ for every truth assignment v and every formula φ , by induction on the length of φ . Write the base case and the cases $\neg, \rightarrow, \leftrightarrow$ in full. (Exercise 2.28(a) is the restriction of this statement to $\{\neg, \wedge, \vee\}$.)
- (b) Show that $(v^*)^* = v$ as *truth assignments*, being explicit about where Theorem 2.2 is used. Deduce that $v \mapsto v^*$ is a bijection from the set of truth assignments to itself.
- (c) Deduce that φ is a tautology if and only if φ^* is a tautology. Write the quantifier argument out in full: given an arbitrary truth assignment w , say exactly which assignment you feed into (a), and why every truth assignment arises in the required form.
- (d) Deduce, without any new induction, that φ is a contradiction if and only if φ^* is a contradiction.
- (e) Prove or refute, without any new induction: $v(\varphi^*) = v^*(\varphi)$ for every truth assignment v and every formula φ .

Problem 10 ($\star$ The substitution lemma). Fix a propositional variable p and a formula χ . For each formula $\varphi \in \text{Form}(P, S)$, $S = \{\neg, \wedge, \vee, \rightarrow, \leftrightarrow\}$, define the formula $\varphi[\chi/p]$ (“ φ with χ substituted for p ”) by recursion:

$$\begin{aligned} p[\chi/p] &= \chi; & q[\chi/p] &= q \text{ for each variable } q \neq p; & (\neg\theta)[\chi/p] &= \neg(\theta[\chi/p]); \\ (\theta \circ \psi)[\chi/p] &= (\theta[\chi/p] \circ \psi[\chi/p]) \text{ for each } \circ \in \{\wedge, \vee, \rightarrow, \leftrightarrow\}. \end{aligned}$$

- (a) Explain in one or two sentences why this recursion specifies $\varphi[\chi/p]$ unambiguously, citing the relevant fact from §2.2.
- (b) Prove: for every truth assignment v ,

$$v(\varphi[\chi/p]) = u(\varphi),$$

where u is the truth assignment determined by $u(p) = v(\chi)$ and $u(q) = v(q)$ for every variable $q \neq p$. (Which result guarantees that such a u exists and is unique?) Use induction on the length of φ ; write the base case — both subcases — and the cases $\neg$ and $\rightarrow$ in full.

- (c) Deduce: if φ is a tautology, then $\varphi[\chi/p]$ is a tautology, for every formula χ .
- (d) Show that the converse of (c) fails, following the refutation discipline.
- (e) In at most three sentences, use (b) to explain the following phenomenon: in a schematic truth table such as the one in Problem 6(b), every row is realizable when φ and ψ are distinct propositional variables, yet rows can become unrealizable once φ and ψ are correlated, as in Problem 6(d).

Part IV — Tautologies, contradictions, counterexamples

Problem 11 (Calibration: true or false?). For each claim, decide whether it is true or false. Give a short proof for each true claim; follow the refutation discipline for each false one. Here u, w, v range over truth assignments and φ, ψ over formulas.

- (a) If $u(\varphi) = w(\varphi)$ for some formula φ , then $u(q) = w(q)$ for every propositional variable q occurring in φ .
- (b) There is a truth assignment v with $v((p \wedge q)) = \text{T}$ and $v(q) = \text{F}$.
- (c) There is a function $f: \text{Form}(P, S) \rightarrow \{\text{T}, \text{F}\}$ with $f((p \wedge q)) = \text{T}$ and $f(q) = \text{F}$.
- (d) If φ is not a tautology, then $\neg\varphi$ is a tautology.
- (e) If φ is not a tautology, then $\neg\varphi$ is not a contradiction.
- (f) If neither φ nor ψ is a tautology, then $(\varphi \vee \psi)$ is not a tautology.
- (g) If $(\varphi \wedge \psi)$ is a tautology, then φ is a tautology.
- (h) If φ is not a contradiction, then some truth assignment makes φ true.

Problem 12 (Around modus ponens: Exercise 2.27 revisited). Let φ, ψ be formulas. Prove or refute each of the following; refutations must follow the refutation discipline.

- (a) If φ and ψ are tautologies, then $(\varphi \wedge \psi)$ is a tautology.
- (b) If $(\varphi \wedge \psi)$ is a tautology, then φ and ψ are both tautologies.
- (c) If $(\varphi \vee \psi)$ is a tautology, then φ is a tautology or ψ is a tautology.
- (d) If $(\varphi \rightarrow \psi)$ is a tautology and ψ is a contradiction, then φ is a contradiction.

- (e) If $(\varphi \leftrightarrow \psi)$ is a tautology and φ is a tautology, then ψ is a tautology.
- (f) If $(\varphi \leftrightarrow \psi)$ is a tautology, then φ and ψ are tautologies.

Problem 13 (Forced, impossible, or independent). An *instance* of a hypothesis H about metavariables φ, ψ is a choice of actual formulas for the metavariables under which H holds. Given H and a conclusion C , exactly one of the following obtains:

- (I) every instance of H satisfies C (“ H forces C ”);
- (II) no instance of H satisfies C (“ H forces the failure of C ”);
- (III) some instance of H satisfies C and some instance of H does not (“ C is independent of H ”).

Note the difference between (II) and the mere failure of (I): in case (III) the conclusion is not forced — it can go either way — whereas in case (II) it is forced to fail.

For each pair below, decide which of (I), (II), (III) obtains. In cases (I) and (II), prove the forcing. In case (III), give *two* fully concrete instances — one where C holds and one where it fails — verifying in each instance every part of H and the status of C , per the refutation discipline.

- (a) H : ψ is a tautology. C : $(\varphi \rightarrow \psi)$ is a tautology.
- (b) H : $(\varphi \rightarrow \psi)$ and ψ are tautologies. C : φ is a tautology.
- (c) H : φ is a contradiction. C : $(\varphi \rightarrow \psi)$ is a tautology.
- (d) H : φ is a contradiction. C : $(\varphi \wedge \psi)$ is a tautology.
- (e) H : $(\varphi \rightarrow \psi)$ is a contradiction. C : ψ is a tautology.
- (f) H : φ is not a tautology. C : $\neg\varphi$ is a tautology.
- (g) Parts (d) and (e) tacitly use the fact that no formula is both a tautology and a contradiction. Prove this fact carefully. Which half of Theorem 2.2 — existence or uniqueness — does your proof use?

Problem 14 (Characterizations: Exercise 2.31 revisited). Let φ, ψ be formulas.

- (a) Suppose φ is a tautology. Under what circumstances, if any, is $(\varphi \rightarrow \psi)$ a tautology? State and prove a characterization of the form “... if and only if ...”.
- (b) Suppose ψ is a tautology. Under what circumstances, if any, is $(\varphi \rightarrow \psi)$ a contradiction? Prove your answer.
- (c) If $(\varphi \wedge \psi)$ is a contradiction, must φ or ψ be a contradiction? Prove or refute.
- (d) Suppose $(\varphi \vee \psi)$ is a contradiction. State and prove the strongest conclusion you can about φ and ψ , in the form of a characterization of when $(\varphi \vee \psi)$ is a contradiction.
- (e) If $(\varphi \leftrightarrow \psi)$ is a contradiction, must φ or ψ be a contradiction? Prove or refute. Then state and prove a characterization: $(\varphi \leftrightarrow \psi)$ is a contradiction if and only if ... (a condition relating the values of φ and ψ under each truth assignment).

Problem 15 (Majority rule: Exercise 2.29 revisited). Let S be a finite set of truth assignments with $|S|$ odd, and let D be the set of formulas made true by more than half of the members of S .

- (a) Show that for every formula φ , exactly one of φ , $\neg\varphi$ belongs to D .
- (b) Show that if $\varphi \in D$ and $(\varphi \rightarrow \theta)$ is a tautology, then $\theta \in D$.
- (c) Construct a set S of three truth assignments on $\{p, q\}$ — specify each assignment by its values on p and q — such that $p \in D$ and $(p \rightarrow q) \in D$ but $q \notin D$. Verify all three memberships.
- (d) If $\varphi \in D$ and $\psi \in D$, must $(\varphi \wedge \psi) \in D$? Prove, or refute with a concrete S and concrete formulas.
- (e) Now drop the requirement that $|S|$ is odd: exhibit a set S with $|S| = 2$ for which the conclusion of (a) fails.
- (f) Show that if $|S| = 1$ then D is closed under modus ponens: if $\varphi \in D$ and $(\varphi \rightarrow \theta) \in D$, then $\theta \in D$. Identify the precise step of your argument that breaks down when $|S| = 3$, and explain how your example in (c) exploits exactly that gap.

Part V — Capstone

Problem 16 ($\star$ Repetition-free formulas: Exercise 2.30 generalized). Call a formula *repetition-free* if no propositional variable occurs more than once in it.

- (a) (*Gluing Lemma.*) Let θ and ψ be formulas with no propositional variable in common, and let u, w be truth assignments. Show that there is a truth assignment x with $x(\theta) = u(\theta)$ and $x(\psi) = w(\psi)$. Define x explicitly on the propositional variables, and cite Theorem 2.2 and Problem 8 at the precise points where each is used.
- (b) Prove by induction on length: every repetition-free formula of $\text{Form}(P, S)$, $S = \{\neg, \wedge, \vee, \rightarrow, \leftrightarrow\}$, is made true by some truth assignment and made false by some truth assignment. A complete proof should record why the immediate subformulas of a repetition-free formula are themselves repetition-free and, in the binary cases, have no variable in common; it should also identify the feature shared by the truth tables of all five connectives that makes the inductive step go through.
- (c) Deduce Exercise 2.30, and state the stronger conclusion that your proof of (b) actually delivers.
- (d) Exhibit a specific formula containing a repeated variable, and point to the specific step of your inductive argument in (b) that fails for it.
- (e) In one short paragraph: the freedom exploited in (a) is exactly what is *not* available in Problem 13(b) (equivalently, Exercise 2.27(c)). Explain why distinct metavariables φ, ψ need not behave like distinct propositional variables, and which conditions on φ and ψ would restore the analogy.